

Laxmi Charitable Trust's
Sheth L.U.J College of Arts & Sir M.V. College of Science and Commerce
Department of Information Technology (B.Sc.I.T Semester V)
Applied Business Analytics

M2 Practical 7

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7. Sampling and Sampling Distributions:

Part A: Survey and Sampling

1. Select a real-life business topic for conducting a survey.

Topic: Social Media Awareness for Bird Rescue and Community-Based Organizations.

2. Prepare a questionnaire containing at least 5 questions related to the selected topic.

A questionnaire containing questions related to bird rescue and social media awareness was prepared.

3. Collect responses from at least 50 people.

Total 45 responses were collected through the survey.

4. Store the collected data in a CSV file or Pandas DataFrame.

5. Display the first five records of the collected data.

Code:

```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
df = pd.read_csv("Bird Rescue Survey .csv")
print("FIRST FIVE RECORDS:")
print(df.head())
```

Output:

FIRST FIVE RECORDS:				
	Timestamp	Name	Age	\
0	9/18/2026 22:05:21	Radha	19	- 25
1	9/18/2026 22:10:56	Lavkush Vishwakarma	19	- 25
2	9/18/2026 22:12:12	Khushi	19	- 25
3	9/18/2026 22:12:31	Priti Yadav	19	- 25
4	9/18/2026 22:33:06	Rani	25	Above

6. Identify the population and sample for the survey.

Code:

```
print("\nTotal Population/Responses:", len(df))
```

Output:

Total Population/Responses: 45

7. Identify the sampling technique used to select the sample.

Code:

```
sample = data.sample(n=30, random_state=42)
```

8. Identify the numerical and categorical variables in the collected data.

9. Select one numerical variable for further analysis.

Code:

```
data = pd.to_numeric(df.iloc[:, 8],
errors="coerce").dropna()
print("Valid observations:", len(data))
```

Output:

Valid observations: 33

10. Calculate the sample size, mean, median, and standard deviation of the selected Variable.

Code:

```
sample_size = len(sample)
sample_mean = sample.mean()
sample_median = sample.median()
sample_std = sample.std()
print("\nPART A - SAMPLE STATISTICS")
print("Sample Size:", sample_size)
print("Sample Mean:", sample_mean)
print("Sample Median:", sample_median)
```

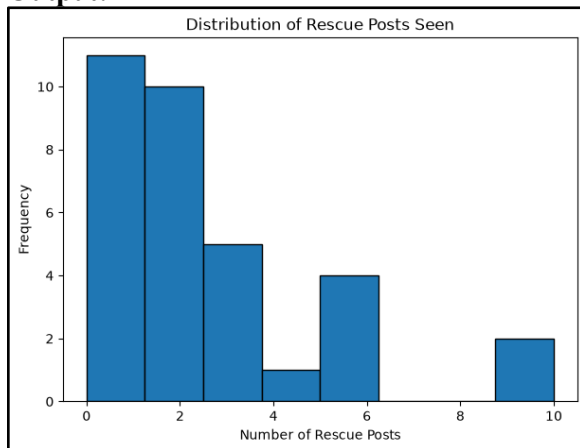
```
print("Sample Standard Deviation:", sample_std)
```

Output:

```
Sample Size: 30
Sample Mean: 2.4
Sample Median: 2.0
Sample Standard Deviation: 2.077797242107695
```

11. Create a suitable chart to display the collected data.**Code:**

```
plt.figure(figsize=(7, 5))
plt.hist(data, bins=8, edgecolor="black")
plt.title("Distribution of Rescue Posts Seen")
plt.xlabel("Number of Rescue Posts")
plt.ylabel("Frequency")
plt.show()
```

Output:**12. Write two observations from the survey data.****Observation 1:**

Most respondents saw a limited number of bird or animal rescue posts on social media.

Observation 2:

The number of rescue posts seen varied among the respondents.

Part B: Sampling Distribution and Central Limit Theorem**13. Select a numerical variable from the collected data.**

Selected numerical variable: Q6 – Number of animal/bird rescue posts seen on social media in the past month.

14. Select a sample of 30 observations randomly from the data.**15. Calculate the mean of the selected sample.****16. Repeat the random sampling process 100 times.****17. Calculate and store the mean of each sample.****Code:**

```
sample_size = 30
number_of_samples = 100
sample_means = []
# Repeat random sampling 100 times
for i in range(number_of_samples):
    random_sample = data.sample(
        n=sample_size,
        replace=True
    )
    mean = random_sample.mean()
    sample_means.append(mean)
```

18. Display the first 10 sample means.**Code:**

```
print("\nFIRST 10 SAMPLE MEANS:")
for i in range(10):
    print("Sample", i + 1, "Mean =", sample_means[i])
```

Output:

```
FIRST 10 SAMPLE MEANS:
Sample 1 Mean = 2.1
Sample 2 Mean = 2.6
Sample 3 Mean = 2.8666666666666667
Sample 4 Mean = 2.7
Sample 5 Mean = 3.0333333333333333
Sample 6 Mean = 2.5666666666666667
Sample 7 Mean = 2.3
Sample 8 Mean = 3.1666666666666665
Sample 9 Mean = 1.8333333333333333
Sample 10 Mean = 2.2333333333333334
```

19. Calculate the mean of all the sample means.**Code:**

```
mean_of_sample_means = np.mean(sample_means)
print("\nMean of All Sample Means:",
      mean_of_sample_means)
# Original population mean
population_mean = data.mean()

print("\nPopulation Mean:",
      population_mean)
```

Output:

```
Mean of All Sample Means: 2.6196666666666664
```

```
Population Mean: 2.606060606060606
```

20. Calculate the standard deviation of the sample means.

Code:

```
std_sampling_distribution = np.std(
    sample_means,
    ddof=1
)
print("Standard Deviation of Sample Means:",
      std_sampling_distribution)
```

Output:

```
Standard Deviation of Sample Means: 0.40416171158752257
```

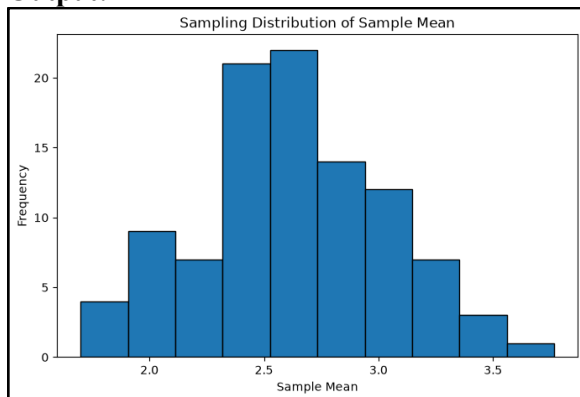
Output:

21. Create a histogram of the sample means.

Code:

```
plt.figure(figsize=(8, 5))
plt.hist(
    sample_means,
    bins=10,
    edgecolor="black"
)
plt.title("Sampling Distribution of Sample Mean")
plt.xlabel("Sample Mean")
plt.ylabel("Frequency")
plt.show()
```

Output:



22. Observe the shape of the sampling distribution.

The sampling distribution is approximately bell-shaped and concentrated around the central value.

23. Compare the original sample mean with the mean of the sampling distribution.

Code:

```
print("\nCOMPARISON")
print("Original Sample Mean:", sample_mean)
print("Mean of Sampling Distribution:",
```

```
      mean_of_sample_means)
print("\nPopulation Mean:", population_mean)
```

Output:

```
COMPARISON
Original Sample Mean: 2.4
Mean of Sampling Distribution: 2.6196666666666664

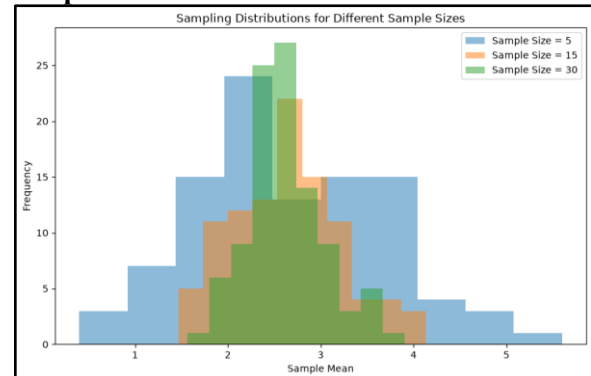
Population Mean: 2.606060606060606
```

24. Compare the sampling distributions obtained for different sample sizes.

Code:

```
sample_sizes = [5, 15, 30]
plt.figure(figsize=(10, 6))
for size in sample_sizes:
    means = []
    for i in range(100):
        random_sample = data.sample(n=size, replace=True)
        means.append(random_sample.mean())
    plt.hist(means, bins=10, alpha=0.5, label=f"Sample Size = {size}")
plt.title("Sampling Distributions for Different Sample Sizes")
plt.xlabel("Sample Mean")
plt.ylabel("Frequency")
plt.legend()
plt.show()
```

Output:



25. Calculate the standard error of the sample mean.

Code:

```
standard_error = data.std(ddof=1) / np.sqrt(30)
print("\nPOINT 25 - STANDARD ERROR")
print("Standard Error of Sample Mean:", standard_error)
```

Output:

```
Standard Error of Sample Mean: 0.43765329204794523
```

26. Explain how the Central Limit Theorem is demonstrated through the experiment.

The Central Limit Theorem was demonstrated using repeated random samples. The distribution of sample means became more concentrated as sample size increased.

27. Use the sample mean to estimate the population mean.

Code:

```
print("\nPOINT 27 - ESTIMATION OF POPULATION  
MEAN")  
estimated_population_mean = sample_mean  
print("Sample Mean:", sample_mean)  
print("Estimated Population Mean:",  
estimated_population_mean)
```

Output:

```
Sample Mean: 2.4  
Estimated Population Mean: 2.4
```

28. Write a conclusion based on the sampling distribution and Central Limit Theorem.

Thus, sampling and sampling distribution were successfully performed using Python. The results demonstrated the Central Limit Theorem and showed that the sample mean can be used to estimate the population mean.